

Predicting Mortgage Backed Securities Prepayment Using Machine Learning Methods

UE24CS352A – Machine Learning | Mini-Project Write-up

Section D | Team D-12 | Project No. 55

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1. Problem Statement and Objective

When a borrower pays off a mortgage early (prepayment), investors in mortgage-backed securities (MBS) stop receiving the scheduled interest on that loan, so predicting prepayment is central to MBS risk. We treat it as loan-level binary classification: for a loan that is still active at the end of month t , will it be paid off in full in month $t+1$?

Original paper. Guo, Boccio and Cameron (Stanford CS229, 2021) labelled each loan-month as “prepaid in this month”, split rows at random and compared logistic regression, SVMs, Gaussian discriminant analysis (GDA) and a neural network. The paper reports its best test accuracy, 99.959%, for quadratic GDA (the paper’s result, not ours).

Our implementation. We reimplement the same model families on Freddie Mac data, show why accuracy at that level is not attainable on a leakage-free version of the task, correct the evaluation (target definition, data split and metrics), and compare all models using metrics suited to a rare positive class.

2. Dataset

- **Source:** Freddie Mac Single-Family Loan-Level Dataset (sample files). The origination file (one row per loan) is joined on loan ID to the monthly performance file (one row per loan per month).
- **Vintages:** 2012, 2014, 2016, 2018, 2020 and 2022; 30-year, fixed-rate, fully amortizing loans; 5,000 loans sampled per vintage, giving 29,982 loans.
- **Size:** 1,722,603 usable loan-month observations (reporting months January 2012 to February 2026) with 104 final features after preprocessing.
- **Macroeconomic data (FRED):** 30-year mortgage rate (MORTGAGE30US), FHFA house price index (HPIPNM226S) and unemployment rate (UNRATE), joined by reporting month.
- **Features:** origination attributes (credit score, LTV, CLTV, DTI, balance, interest rate, occupancy, purpose, state, etc.), loan history (age, remaining term, balance ratio, delinquency, modification) and engineered market features: refinance incentive (loan rate minus market rate), burnout, mark-to-market LTV, house-price growth, unemployment change and seasonality.
- **Target and class balance:** 1 if the loan’s next monthly record shows zero-balance code 01 (voluntary payoff), excluding maturity. Only 1.08% of observations are positive (18,556 prepayments vs. 1,704,047 non-prepayments).

3. Methodology

- **Next-month target:** features use only information known at the end of month t . The payoff row itself, whose reported balance is already 0, is never used as a feature row.
- **Loan-level 60/20/20 split** (seed 42): train 1,031,599 rows (17,989 loans), validation 344,513 rows (5,996 loans), test 346,491 rows (5,997 loans). No loan appears in more than one set.
- **Preprocessing fitted on training data only:** missing-value codes (e.g. credit score 9999, LTV/DTI 999) set to missing; median imputation plus a missing-value indicator; one-hot encoding of 9 categorical variables (levels with at least 50 training rows); standardization; constant columns dropped.
- **Validation-based threshold:** each model’s decision threshold is the one that maximizes F1 on the validation set; it is then applied unchanged to the test set.
- **Implementation:** Python with pandas, scikit-learn and PyTorch. `run_all.py` builds the dataset, trains, evaluates and saves all models; `demo.py` scores held-out test loan-months with the saved models without retraining.

Problem in original setup	Our correction
Same-month label leakage	Next-month target
Random row split	Loan-level split
Accuracy on highly imbalanced data	PR-AUC, precision, recall and validation threshold

Table 1. Methodological corrections relative to the original paper’s setup.

4. Machine Learning Models

- **Logistic regression:** no penalty, L1 and L2 ($C = 0.1$); trained on a 600,000-row training subsample.
- **SVM:** polynomial (degree 3), RBF and sigmoid kernels ($C = 1$); 15,000-row training subsample (kernel cost).
- **GDA:** linear (LDA) and quadratic (QDA; regularization 0.001 selected on validation).
- **Feed-forward neural network (PyTorch):** 104–128–256–1 with batch normalization and ReLU, binary cross-entropy with positive-class weight 10, SGD (learning rate 0.1, $\times 0.91$ per epoch), 30 epochs; the epoch with the best validation PR-AUC is kept.
- **Reference models:** an always-“no” baseline, and gradient boosting (HistGradientBoosting), which we added as a benchmark; *gradient boosting was not in the original paper*.

5. Results (OUR RESULTS)

Model	Accuracy	PR-AUC	ROC-AUC
Gradient Boosting†	96.11%	0.0382	0.7557
Neural Network	96.96%	0.0380	0.7393
Logistic L1	94.80%	0.0314	0.7271
Logistic L2	94.77%	0.0313	0.7272
Logistic No Penalty	94.79%	0.0313	0.7272
GDA Linear	93.96%	0.0300	0.7210
GDA Quadratic	92.14%	0.0210	0.6603
SVM RBF	90.37%	0.0170	0.6093
SVM Polynomial	89.60%	0.0151	0.5759
SVM Sigmoid	90.63%	0.0124	0.5437
Baseline (always “no”)	98.93%	0.0107	0.5000

Table 2. OUR RESULTS on the held-out test set (346,491 loan-months, 3,705 prepayments), sorted by PR-AUC. Random guessing gives PR-AUC 0.0107 (the positive rate). † Benchmark, not in the original paper. The baseline is not a useful model: its accuracy comes from always predicting the majority class and it detects no prepayments.

6. Leakage Analysis

Setup (L2 logistic regression, same loans)	Accuracy	Precision	Recall	PR-AUC
Paper setup reproduced (OUR RESULTS)	99.95%	0.972	0.985	0.987
Fixed setup (OUR RESULTS)	94.77%	0.045	0.193	0.031

Table 3. Effect of the original setup versus the corrected setup.

The paper setup labels each row “prepaid this month”, keeps the payoff row and splits rows at random. Freddie Mac reports the current balance as 0 in the payoff month, so the balance feature effectively reveals the label, and months of the same loan appear in both training and test data. On our data this setup gives 99.95% accuracy, close to the paper’s 99.951% for L2 logistic regression (the paper’s result). With a next-month target and a loan-level split, PR-AUC falls from 0.987 to 0.031: the near-perfect scores come from leakage, not predictive skill. The comparison changes target, balance feature and split together, not each one separately.

7. Key Findings

- Accuracy is misleading with a 1.08% positive class: the always-“no” baseline has the highest accuracy (98.93%) yet detects zero prepayments.
- Gradient boosting (PR-AUC 0.0382) and the neural network (0.0380) are strongest, about $3.6\times$ random (0.0107).
- Logistic L1, L2 and no-penalty results are very similar (PR-AUC 0.0313–0.0314; ROC-AUC 0.7271–0.7272).
- QDA, the paper’s best model, does not reproduce its strong result: PR-AUC 0.0210, below LDA (0.0300).
- SVMs have the lowest PR-AUC (0.0124–0.0170); they were trained on a 15,000-row subsample.
- Largest L1 logistic coefficients: rate incentive (+0.417), mortgage rate (−0.392), mark-to-market LTV (−0.341).

8. Conclusion and Limitations

The original paper’s accuracy of about 99.9% reflects its problem setup rather than predictive skill: reproducing it on our data gives 99.95% accuracy, while the leakage-free, loan-level evaluation gives far lower but realistic scores. Gradient boosting and the neural network perform best (PR-AUC about 0.038), followed by the logistic models and LDA. For a rare event like monthly prepayment, PR-AUC, precision and recall should be reported instead of accuracy.

Limitations: Freddie Mac sample data (5,000 loans per vintage); SVM and logistic models trained on subsamples; national rather than regional macroeconomic variables; no out-of-time evaluation was run (training and test sets share calendar months); results come from a single experimental run (seed 42).

References

- [1] C. Guo, J. Boccio, C. Cameron, “Predicting Mortgage Backed Securities Prepayment Using Machine Learning Methods,” Stanford CS229 project report, Spring 2021. <https://cs229.stanford.edu/proj2021spr/report2/81995903.pdf>
- [2] Freddie Mac, Single-Family Loan-Level Dataset and General User Guide. <https://www.freddiemac.com/research/datasets/sf-loanlevel-dataset>
- [3] Federal Reserve Bank of St. Louis, FRED Economic Data: MORTGAGE30US, HPIPNM226S, UNRATE. <https://fred.stlouisfed.org/>
- Code: <https://github.com/PRANAVKALLA230106/Securities-Prepayment-Using-Machine-Learning-Methods>